

M13-C3 — H4b cautious split under leading a target c on the H3 fixture

H4b run / analysed / not a Bucket 3 claim

1 Status

H4b run / analysed. Bounded follow-up to H3/H4: repository `baseline_cautious` vs locked H3 ECAI brave on `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, focus target c . Markdown twin: `h4b_cautious_split_under_a.md`.

Lead (Layer B): cautious **only** changes the α_3 close — rejects brave residual `c_alpha_3 :- alpha_1, a_val_1` and accepts `c_alpha_3 :- d_val_1`; both arms **solved**. **Shared (Layer A — not a cautious effect):** both arms still gate on BK-leading a and keep the α_1 -nest vs α_2 -flat contrary split. **Not H5.** H0–H4 closed. **Not a Bucket 3 claim.**

2 Research question

On the H3 BD AND + BK-leading- a table, under otherwise shared `nd/re/to` settings, does switching from ECAI brave to repository-baseline cautious change how the split beneath the a gates is closed — in particular the attack on `alpha_3` — and what stays the same?

3 Main finding (Layer B — lead)

Both arms **solve**. The string difference is **exactly one rule** — the body of `c_alpha_3`:

```
% brave:    c_alpha_3(A) :- alpha_1(A), a_val_1(A).
% cautious: c_alpha_3(A) :- d_val_1(A).
```

Decisive cautious fork:

```
folding result: c_alpha_3(A) <- [a_val_1(A)]
  found: alpha_1/1 ... K0, cannot introduce an assumption for [a_val_1(A)]
folding result: c_alpha_3(A) <- [d_val_1(A)]
```

Locked brave at the same fold: `alpha_1 OK` $\rightarrow$ `c_alpha_3 :- alpha_1, a_val_1` (never folds d_* here).

Lead: `baseline_cautious` keeps H3's leading- a gates and the persistent α_1 -nest vs α_2 -flat contrary split; it **only** changes the α_3 close. Both arms **solved**.

Versus H4: H4 blocked a brave residual gadget on **roots** $\rightarrow$ `completed_no_solution`; H4b blocks *this* residual gadget on **child** c $\rightarrow$ still **solved**, with explicit `d_val_1`.

4 Shared background (Layer A — not cautious)

Both arms still gate on BK-leading a ($c \text{ :- } \alpha_1, a_val_1$ / $c \text{ :- } \alpha_2, a_val_0$); evaluator BD AND is not recovered; body vars $\{a\}$.

Contrary skeletons (identical through α_3 intro):

Gate	Contrary structure (both arms)
α_1 ($a=1$)	Nested: $c_alpha_1 \text{ :- } b_val_0$ plus $c_alpha_1 \text{ :- } \alpha_3, b_val_1$
α_2 ($a=0$)	Flat: $c_alpha_2 \text{ :- } d_val_0$ and $c_alpha_2 \text{ :- } b_val_0$

Procedural reason (search — not cautious): c_alpha_1 repaired first; under $a=1$ commit b_val_0 then mint α_3 on residuals; under $a=0$, **relto** KO on existing α_3 forces flat d/b cover. Do **not** credit cautious with symmetrising / removing the nest / fixing the asymmetry.

5 Brave vs cautious table

Layer A objects are the same; Layer B differs only in c_alpha_3 (α_1, a_val_1 vs d_val_1). Both outcomes **solved**.

6 Setup / paths

Same H3 fixture/sample (sha256 48bd0da0...db05). Intervention: locked ECAI brave $\rightarrow$ **baseline_cautious** (not H5). Incidental: collection targets $a/b/d$ timed out at 300s — outside focus.

7 Semantic boundary

Authoritative: Prolog learner outcome under selected **learning_mode**. Under cautious learning, SAT of **bk.sol_chk.asp** is an existential integrity witness only.

8 Boundaries / non-claims

Not a Bucket 3 claim; not H5; not mechanism recovery. Layer A is shared with H3 and is not a cautious effect. H5 and H6 analysed separately (**h5_greedy_cautious.md**, **h6_folding_and_n_ablation.md**); H7a analysed separately (**h7a_relto_vs_sechk.md**); H7b not started.

9 Next

H4b documentation is complete. H5 is run / analysed (**h5_greedy_cautious.md**). H6 is analysed (**h6_folding_and_n_ablation.md**); H7a analysed (**h7a_relto_vs_sechk.md**); H7b not started. Still **no Bucket 3 claim**.